

JOYDEB MONDAL

Principal Applied Scientist

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PROFESSIONAL SUMMARY

Principal Applied Scientist with ~10 years architecting and shipping production AI systems at scale, spanning foundation model fine-tuning and serving, GenAI safety infrastructure, and privacy-preserving enterprise platforms. Technical roadmap owner, cross-team architecture lead, and mentor to 4 Applied Scientists at Oracle Cloud; inventor on multiple AI patents; published at ACL (2022, 2025) and NeurIPS (2024). Technical depth spans production GenAI systems, including PEFT-based adaptation, retrieval-augmented generation(RAG), agentic orchestration, safety evaluation, and scalable inference infrastructure.

EXPERIENCE

Oracle Cloud Infrastructure

2020 – Present

Principal Applied Scientist — GenAI Safety, Foundation Models & Healthcare AI

GenAI Safety & Responsible AI Infrastructure

- Defined and governed the Responsible AI platform strategy for OCI, establishing enterprise-wide AI governance frameworks, safety design standards, and evaluation benchmarks adopted across all production architectures and 15+ enterprise customers.
- Engineered and shipped a production AI guardrails safety platform across OCI's core and customer-facing GenAI services, with integrated PII detection, prompt injection defense, content moderation, and off-topic detection with fine-tuned LLMs and SLMs. Reduced unsafe prompt success rates by 35–50% and AI policy violations by 40%, directly unblocking enterprise customer onboarding in regulated environments.
- Architected a production grade multi agent guardrail platform where specialized agents, sub agents, and safety tools including PII Detection, Content Moderation, Prompt Injection Defense, and Off Topic Classification collaboratively govern GenAI interactions through centralized policy orchestration. Built an extensible agent and tool framework enabling independent safety agents and sub agents to enforce explainable Responsible AI policies, auditability, privacy protection, and harmlessness alignment across enterprise AI systems.
- Standardized an Automatic Prompt Optimization (APO) framework, safety evaluation methodology, and red team evaluation protocols for systematic adversarial stress-testing of safety model robustness, with a unified evaluation output format across all safety models.
- Established a hybrid SLM-first architecture with risk-scored LLM escalation, sustaining sub-100ms p99 latency at multi-million requests per day across safety-critical GenAI endpoints.

Foundation Model Training, Serving & Inference

- Engineered a unified training, evaluation, and inference framework for encoder-based SLMs covering token and text classification tasks, including NER, Aspect-Based Sentiment Analysis, text classification, and sentence sentiment. The shared architecture standardized model development workflows and enabled consistent multi-task benchmarking across OCI Language services.
- Standardized scalable multi model serving infrastructure using NVIDIA Triton, TensorFlow Serving, and OpenVINO, optimizing GPU utilization and inference scheduling for production SLM workloads. Improved serving efficiency with 2 to 3x higher throughput and 30 to 40 percent lower p99 latency compared to legacy single model deployments..
- Implemented PEFT-based distributed fine-tuning using LoRA and QLoRA with Accelerate and DeepSpeed, reducing model adaptation cost by 40–60% with competitive task performance. Delivered vLLM-based LLM serving stacks achieving 3–5x higher token throughput. Integrated hybrid RAG pipelines combining BM25 and dense retrieval with sub-100ms production latency and horizontal scalability.

Healthcare AI Platform

- Engineered and deployed four Medical BERT-based clinical NLP models across enterprise EHR/EMR pipelines: Health NER (47 entity types, CRF decoding, 0.96 macro-F1 across open-source and internal clinical benchmarks), Assertion Detection (5 clinical dimensions: Certainty, Temporality, Subject, Course, Severity), Entity Linking (bi-encoder FAISS retrieval across 670K+ concepts in ICD-10/11, RxNorm, and SNOMED CT with cross-encoder re-ranking), and Relation Extraction (entity marker tokens, directed span-pair classification). All models use a domain-adapted Medical BERT encoder pre-trained on clinical corpora (PubMed, MIMIC-III, and clinical trial documents), reducing clinical term UNK rate from ~10% to under 2%.
- Architected a HIPAA/GDPR-compliant PHI detection model using GLiNER architecture with a context-aware re-lexicalization framework for de-identification workflows, extending the safety alignment and model governance principles of the Responsible AI platform into regulated clinical environments. Confidence-gated LLM augmentation improved entity recall on complex clinical text by 20–25% over the baseline encoder-only approach.

- Directed the evolution from rule-based intent and entity systems to Transformer-based NLP pipelines, improving intent classification accuracy by 25–30% and reducing fallback responses by 20%.
- Engineered and deployed LSTM-based NER models and contextual response prediction systems for enterprise service automation, supporting dialogue-to-solution mapping across multiple verticals.
- Extended the platform into voice-enabled systems through enterprise STT integration and phoneme-aware correction models, improving domain-specific speech recognition accuracy by 15–20%.
- Built predictive analytics modules for HR decision workflows covering attrition, salary estimation, and performance modeling. Mentored junior engineers in applied NLP and conversational AI.

- Established early expertise in production safety systems through an ML-based network anomaly detection system using Random Forest (F1: 92%) on structured traffic features, identifying suspicious behavior in high-volume network environments.

PUBLICATIONS & PATENTS

- "ExpertNeurons at SciVQA-2025: Retrieval Augmented VQA with VLM (RAVQA-VLM)" — Mondal, Bhat, Sarkar — **ACL** 2025
- "NeurIPS 2024 LLM Privacy — ExpertNeurons" — Bhat, Mondal, Sarkar — **NeurIPS** 2024
- "ExpertNeurons at FinCausal 2022: Causality Extraction for Financial Documents" — Mondal, Bhat, Sarkar, Reza. — **ACL** 2022
- "Modgenix at SemEval-2025 Task 1: CAViLR for Multimodal Idiomaticity Understanding" — Mondal, Sarkar — **SemEval** 2025
- "A framework for effective stress testing and application parameter prediction" — **Patent** US 2024/0103925
- "Knowledge based re-lexicalization of PII/PHI entities to enhance data security" — **Patent (filed)** IDF-139790
- "Context aware topic management for multi-turn interactions across dynamic topic sets"— **Patent (filed)** IDF-141488

TECHNICAL SKILLS

AI Systems & Research: LLM/SLM fine-tuning, PEFT (LoRA, QLoRA), safety alignment, responsible AI governance, model behavior evaluation, red teaming, adversarial evaluation, safety evaluation, Constitutional AI principles, agentic systems, representation learning, RAG, prompt and context engineering, APO, LLMOps, evaluation methodology

ML Frameworks: PyTorch, TensorFlow, Scikit-learn, DeepSpeed, Accelerate, Hugging Face Transformers

Agentic and Orchestration: LangGraph, LangChain, LlamaIndex, CrewAI, LangSmith, Model Context Protocol (MCP), A2A, OpenAI API, Anthropic API

Serving and Inference: NVIDIA Triton, vLLM, TensorFlow Serving, Ollama, OpenVINO, inference optimization, GPU utilization, kernel optimization, distributed training, Kubernetes, Docker, CI/CD

Search and Retrieval: Hybrid search (BM25 + dense retrieval), FAISS, Qdrant, ScaNN, OpenSearch, bi-encoder and cross-encoder retrieval

Platforms and Engineering: Python, SQL, Pandas, OCI, AWS, GCP (Vertex AI), Azure ML, FastAPI, Flask, Monitoring and Governance

EDUCATION

MTech, Computer Science — IIT (ISM) Dhanbad	CGPA: 8.37 / 10 2015–2017
BTech, Information Technology — West Bengal University of Technology	CGPA: 8.14 / 10 2010–2014

AWARDS & RECOGNITION

- Oracle Cloud | 2020 – Present | **Promotions:** Applied Scientist > Senior Applied Scientist > Principal Applied Scientist (2 promotions)
- Edison Award:** Org-wide recognition for leading ML innovation in OCI AI Services
 - Ninja Award:** Top contributor for research output and patent filings across OCI AI
- Nouveau Labs | 2017 – 2020 | **Promotions:** Research Scientist > Senior Research Scientist (1 promotion)
- Nouveau Hero Award:** Company-wide recognition as best performer, contributor, and mentor

CERTIFICATIONS

- [Building AI Agents and Agentic Workflows Specialization](#) | IBM
- [Agentic AI with LangChain and LangGraph](#) | IBM
- [Advanced RAG with Vector Databases and Retrievers](#) | IBM
- [AI Agents and Agentic AI with Python and Generative AI](#) | Vanderbilt University (Coursera)